

Uncertainty-Aware Scientific Machine Learning for Engineering Systems

Modern engineering increasingly depends on computational models that must remain trustworthy when physical conditions vary, measurements are noisy or incomplete, and high-fidelity simulation is too expensive for repeated decision making. My research asks a central question: how can we build models that infer hidden physical quantities, discover governing structure, and support efficient engineering prediction while representing uncertainty honestly? I address this question by combining Bayesian inverse modeling, stochastic physics discovery, operator learning, causal reasoning, and engineering simulation. Advanced manufacturing is an important proving ground because it concentrates many of the challenges that motivate my methods, but the underlying ideas transfer naturally to materials, semiconductor, thermal, energy, and other engineered systems.

Research foundation

Bayesian and stochastic inverse modeling

Inverse problems ask what physical state, parameter, or mechanism produced the measurements we can observe. My work on the inverse heat-conduction problem developed a Bayesian spatiotemporal approach that integrates heat-transfer structure with statistical inference to recover unobserved thermal quantities from sparse measurements while quantifying uncertainty. The work first appeared as a peer-reviewed ASME conference paper in 2023 and, following an editorial recommendation for journal development, was expanded and published in the *ASME Journal of Heat and Mass Transfer* in 2024. I have extended the same principle to instrumented indentation, where stochastic data-consistent inversion is used to recover distributions of elastoplastic properties from load-displacement information while accounting for measurement noise and manufacturing-induced variability. The current extension is being developed with collaborators at NIST and Oak Ridge National Laboratory.

Physics discovery under uncertainty

Parameter estimation is insufficient when the assumed governing model is itself incomplete. To address this problem, I developed the stochastic inverse physics-discovery framework reported in *Communications Physics* in 2026. The method treats unknown coefficients in candidate governing equations as random variables and infers distributions whose push-forward is consistent with the distribution of observed system behavior. Across ten scenarios spanning nonlinear oscillators, chaotic systems, ecological data, and infiltration dynamics, the framework reduced coefficient error by about 80% relative to seven comparison methods and achieved exact structural identification in seven of ten cases. This work reframes governing-equation discovery as an uncertainty-aware inverse problem rather than a noise-sensitive curve-fitting exercise. A complementary direction investigates counterfactual explanations and causal diagnostics for distinguishing dynamically meaningful interactions from spurious candidate terms.

Scientific machine learning for efficient computation

Many engineering workflows remain too slow for repeated inference, optimization, or control because every candidate decision requires another high-fidelity simulation. My CRONet work addresses this bottleneck using a convolutional recurrent operator-approximator network for topology optimization. Published in *Manufacturing Letters*, CRONet learns the evolution of design fields and reduces the computational burden of repeated finite-element calculations while preserving close agreement with benchmark solutions. The work was later extended through collaborative research on hardware acceleration using AMD Versal AIE-ML engines, published at IEEE FCCM 2026. A complementary advanced-manufacturing project develops an image-based 3D reconstruction and surface-inspection system for additively manufactured components. The system combines multi-view photogrammetry for global geometry and scale with photometric stereo for fine surface reconstruction under controlled illumination, followed by comparison with CAD or other reference geometry. Current work focuses on automated image and illumination capture, calibration,

robust object extraction, scale recovery, multi-view fusion, and geometric validation. The initial defect-detection scope emphasizes positive surface deviations such as protrusions and excess material. The longer-term objective is integration with a Meltio robotic metal-additive-manufacturing platform for post-print inspection, in-situ quality monitoring, and targeted surface-processing decisions.

Research areas

1. Uncertainty-aware inverse modeling

I develop probabilistic methods for recovering latent states, physical parameters, and distributions of engineering properties from indirect and heterogeneous observations. A continuing goal is to separate intrinsic physical variability from measurement error, numerical approximation, and model discrepancy rather than allowing these sources to collapse into a single uncertainty term. Hierarchical Bayesian models, data-consistent inversion, Gaussian-process or neural-operator surrogates, and sequential inference provide a methodological toolkit for this work. These methods are especially useful in manufacturing and materials systems where thermal, force, acoustic, image, and geometric data provide partial views of an evolving physical state.

2. Governing-physics and causal discovery

When the model form is not fully known, I study how sparse and interpretable governing relationships can be discovered from uncertain observations. My current work combines stochastic physics discovery with counterfactual explanations and complementary causal diagnostics. The broader goal is to move beyond fitting an equation that predicts well toward determining whether the recovered interactions are physically meaningful, stable under perturbation, and sufficient to explain new observations. This direction creates a natural connection among dynamical systems, scientific machine learning, causal inference, and model-form uncertainty.

3. Scientific machine learning and computational acceleration

I develop surrogate and operator-learning methods that reduce the cost of high-fidelity simulation and make physically grounded models usable inside repeated design, optimization, inverse, and decision workflows. My work spans topology optimization, computing-aware implementation, and image-based additive-manufacturing quality assurance, where controlled imaging, geometric reconstruction, and reference-model comparison provide a pathway from sensing to inspection and process decisions. A central concern is not simply speed, but trust: learned or reconstructed representations should include diagnostics that indicate when a result can be accepted, corrected with additional information, or rejected outside the model's reliable operating regime.

Current and emerging directions

My future work will connect these research areas into adaptive models that can update from new data, recognize when their assumptions are inadequate, and guide the next informative measurement, simulation, or experiment. Three directions are especially promising:

- Uncertainty-aware digital representations that combine inverse inference, physics-based simulation, surrogate prediction, and sequential updating as new observations arrive.
- Self-correcting models that use discrepancy detection and physics discovery to determine whether new behavior reflects parameter drift, sensing error, intrinsic variability, or missing governing structure.
- Active sensing and experimentation that connect uncertainty to value-of-information criteria so that new data are collected where they are most useful for model discrimination and engineering decisions.

Autonomous material characterization provides a natural early testbed because it links sensing, simulation, material-property inference, model adequacy, and experimental selection in one workflow. The same ideas can extend to thermal systems, semiconductor processes, energy systems, and other cyber-physical settings in which a model must remain useful as operating conditions change.

Research culture, collaboration, and impact

I plan to build a research group in which reproducibility and mentorship are treated as technical infrastructure. Version-controlled software, documented benchmarks, reproducible computational workflows, and FAIR data practices will make results auditable and lower the barrier for new students to contribute. Students will begin with tractable modules, present results early, and progressively assume ownership of methodological questions. My existing collaborations with researchers at NIST and Oak Ridge National Laboratory provide a foundation for national-laboratory partnerships, while manufacturing, semiconductor, materials, thermal, and energy applications create opportunities for industry collaboration and externally funded interdisciplinary research.

The broader vision is a class of engineering models that remain useful not because they are assumed to be correct, but because they can quantify what they do not know, identify when their assumptions fail, learn what physics is missing, and update predictions or decisions accordingly. By advancing the statistical and computational foundations of this capability, I aim to contribute methods that are rigorous enough for scientific discovery and practical enough for consequential engineering systems.